

Lecture 2: Univariate Population Inference

August 18, 2023

Review

- ▶ Last lecture, we discussed description/summarization with means, conditional means, and differences in conditional means.
- ▶ We considered the properties of these summaries (e.g., balance point, least squares, etc.) and how much was left out by summarizing (e.g., squared residuals)
- ▶ This was done in the context where we only cared about the sample, either provisionally or because we had sampled everyone (i.e., a census).

Today

- ▶ Today we consider the context where we have a random sample from a large population.
- ▶ Our goal over the next couple of lectures is to summarize the population with population means, conditional population means, and differences in conditional population means.
- ▶ Summarizing data we don't have (everything in the population that isn't in the sample) is more difficult than summarizing data we do have (sample).

Outline

Population Inference for Proportions

Population Inference for Means

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Population Inference for Means

Sample

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Table: Sample

i	1	2	3	4	...	625	$\bar{y}_{625}$
y_i	1	1	0	1	...	0	.66

Sample

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Suppose also that all potential voters can be contacted, will respond honestly to your questions, and will not change their minds about voting.

Table: Sample

i	1	2	3	4	...	625	$\bar{y}_{625}$
y_i	1	1	0	1	...	0	.66

Remember that the mean of a binary variable is a proportion.

Sample versus Population

Table: Sample

i	1	2	3	4	...	625	$\bar{y}_{625}$
y_i	1	1	0	1	...	0	.68

Sample versus Population

Table: Sample

i	1	2	3	4	...	625	$\bar{y}_{625}$
y_i	1	1	0	1	...	0	.68

After election day, we found that in fact 4,999,960 people, or roughly 70% of registered voters, turned out to vote.

Table: Population

j	1	2	3	4	...	...	...	...	7,233,584	$\bar{y}_{7,233,584}$
y_j	0	1	0	1	...	...	...	...	1	.70

Two Questions

1. .68 is close to .70, did we get lucky?
2. Could we have predicted how close we would get before seeing the .70?

Sampling Distribution of $\bar{Y}_{625}$

i	J_1	Y_1	J_2	Y_2	$\dots$	J_{625}	Y_{625}	$\bar{Y}_{625}$
	9562350	1	8763351	1	$\dots$	1294801	0	.68

Sampling Distribution of $\bar{Y}_{625}$

i	1	2	...	625	
s	J_1	J_2	J_{625}	Y_{625}	$\bar{Y}_{625}$
1	9562350	8763351	1294801	0	.68
2	5331704	4533839	3342359	1	.70

Sampling Distribution of $\bar{Y}_{625}$

i	1		2		...	625		$\bar{Y}_{625}$
s	J_1	Y_1	J_2	Y_2	...	J_{625}	Y_{625}	
1	9562350	1	8763351	1	...	1294801	0	.68
2	5331704	0	4533839	1	...	3342359	1	.70
3	5129936	0	10981600	0	...	4096184	1	.75
4	803605	0	7036389	1	...	803605	0	.73
5	148567	0	3833847	1	...	4769869	1	.69
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
1 mil	4163458	0	8384613	1	...	377981	1	.74
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮

Sampling Distribution of $\bar{Y}_{625}$

i	1		2		...	625		$\bar{Y}_{625}$
s	J_1	Y_1	J_2	Y_2	...	J_{625}	Y_{625}	
1	9562350	1	8763351	1	...	1294801	0	.68
2	5331704	0	4533839	1	...	3342359	1	.70
3	5129936	0	10981600	0	...	4096184	1	.75
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⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
$f(\cdot)$								

- What is the distribution of Y_1 ?

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⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
$f.(.)$	Bern(.70)		Bern(.70)			Bern(.70)		

- ▶ What is the distribution of Y_1 ?
- ▶ What is the distribution of $\sum_i^{625} Y_i$?

Sampling Distribution of $\bar{Y}_{625}$

i	1		2		...	625		$\bar{Y}_{625}$
s	J_1	Y_1	J_2	Y_2	...	J_{625}	Y_{625}	
1	9562350	1	8763351	1	...	1294801	0	.68
2	5331704	0	4533839	1	...	3342359	1	.70
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⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
$f(\cdot)$	Bern(.70)		Bern(.70)			Bern(.70)		$\frac{Bin(625, .70)}{625}$

- ▶ What is the distribution of Y_1 ?
- ▶ What is the distribution of $\sum_i^{625} Y_i$?
- ▶ What two numbers show that we sampled with replacement?

Review of Bernoulli and Binomial Distributions

We can write distributions for our random variables (RVs) that describe the probability of getting particular values for these variables.

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$Y_1, \dots, Y_{625}$ are Bernoulli RVs:

$$f_{Y_1}(y_1) = (.71)^{y_1} (.29)^{1-y_1} \text{ for } y_1 = 0, 1$$

$$\vdots = \vdots$$

$$f_{Y_{625}}(y_{625}) = (.71)^{y_{625}} (.29)^{1-y_{625}} \text{ for } y_{625} = 0, 1$$

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$Y_1, \dots, Y_{625}$ are Bernoulli RVs:

$$\begin{aligned} f_{Y_1}(y_1) &= (.71)^{y_1} (.29)^{1-y_1} \text{ for } y_1 = 0, 1 \\ \vdots &= \vdots \\ f_{Y_{625}}(y_{625}) &= (.71)^{y_{625}} (.29)^{1-y_{625}} \text{ for } y_{625} = 0, 1 \end{aligned}$$

$\sum_{i=1}^{625} Y_i$ is a Binomial RV:

$$f_{\sum_{i=1}^{625} Y_i} \left(\sum_{i=1}^{625} y_i \right) = \binom{625}{k} (.70)^{\sum_{i=1}^{625} y_i} (.30)^{625 - \sum_{i=1}^{625} y_i} \text{ for } \sum_{i=1}^{625} y_i = 0, \dots, 625$$

The Sampling Distribution in $\mathbb{R}$

```
# Resample the number of voters 1,000,000  
# times and store these 1,000,000  
# numbers in a vector.
```

The Sampling Distribution in $\mathbb{R}$

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sumY_vec <- rbinom(1000000, size=625, prob=.70)
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The Sampling Distribution in R

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# Divide all of these numbers  
# by the sample size.  
  
Ybar_vec <- sumY_vec/625
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# times and store these 1,000,000
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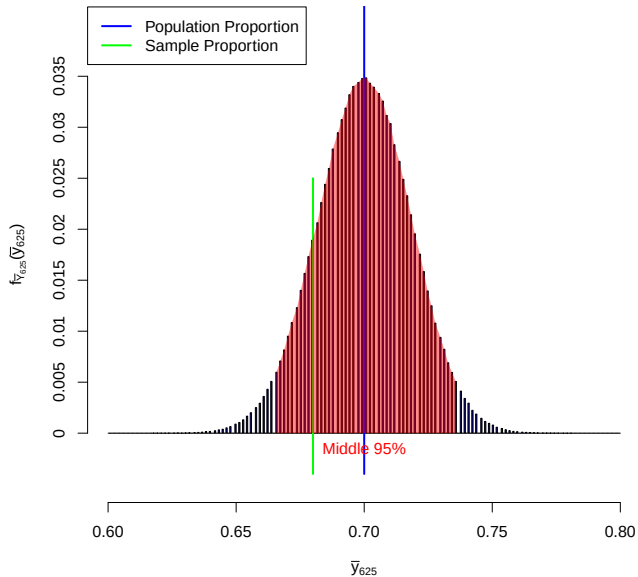
# Divide all of these numbers
# by the sample size.

Ybar_vec <- sumY_vec/625

# Plot a histogram

hist(Ybar_vec)
```

Sampling Distribution of $\bar{Y}_{625}$



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Bootstrapped Sampling Distributions

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However, we can take repeated random samples *with replacement* of size 625 from the sample of size 625.

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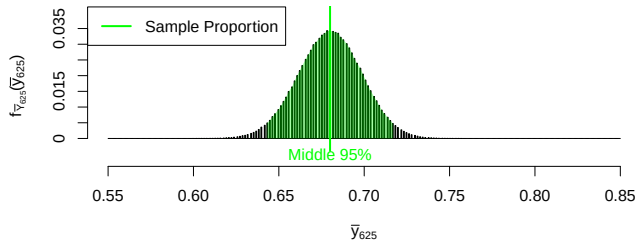
Table: Sample

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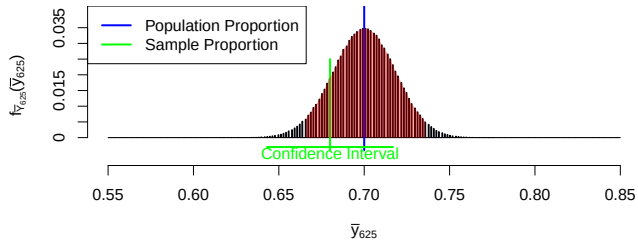
This is equivalent to replacing .70 with .68 in the R code.

```
sumY_vec <- rbinom(1000000, size=625, prob=.68)
```


Estimated Sampling Distribution of $\bar{Y}_{625}$



Sampling Distribution of $\bar{Y}_{625}$



The Logic of Confidence Intervals

If the bootstrapped sampling distribution has the same shape as the sampling distribution,

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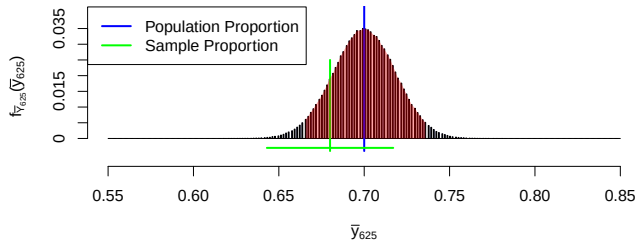
1. The 95% confidence interval will *cover* the population proportion when the sample proportion is in the middle 95% of the sampling distribution.
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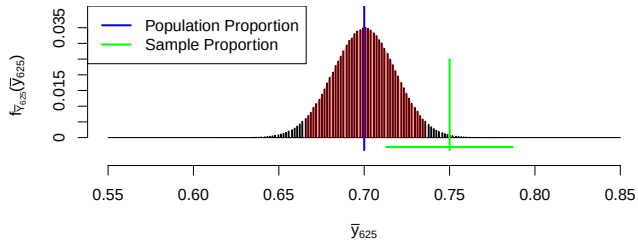
If the bootstrapped sampling distribution has the same shape as the sampling distribution,

1. The 95% confidence interval will *cover* the population proportion when the sample proportion is in the middle 95% of the sampling distribution.
2. The 95% confidence interval will *not* cover the population proportion when the sample proportion is not in the middle 95% of the sampling distribution.
3. There is a 95% probability of getting a sample proportion in the middle 95% of the sampling distribution, therefore there is 95% probability of getting a 95% confidence interval that covers the population proportion.

Sampling Distribution of $\bar{Y}_{625}$



Sampling Distribution of $\bar{Y}_{625}$



The Sampling Distribution of the Confidence Interval

i	1	2	...	625		
s	Y_1	Y_2	...	Y_{625}	$\bar{Y}_{625}$	95% CI
1	1	1	...	0	.68	
2	0	0	...	1	.75	
3	0	1	...	1	.70	
4	0	1	...	1	.73	
5	0	1	...	1	.69	
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	
1 mil	0	1	...	1	.74	
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	

Table: Sampling Distributions for $\bar{Y}_{625}$ and the 95% Confidence Interval

95% of the rows in this sampling distribution will have a CI that covers .70.

The Sampling Distribution of the Confidence Interval

i	1	2	...	625	$\bar{Y}_{625}$	95% CI
s	Y_1	Y_2	...	Y_{625}		
1	1	1	...	0	.68	[.643,.717]
2	0	0	...	1	.75	
3	0	1	...	1	.70	
4	0	1	...	1	.73	
5	0	1	...	1	.69	
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	
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$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	

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1	1	1	...	0	.68	[.643,.717]
2	0	0	...	1	.75	[.715,.785]
3	0	1	...	1	.70	
4	0	1	...	1	.73	
5	0	1	...	1	.69	
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	
1 mil	0	1	...	1	.74	
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	

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3	0	1	...	1	.70	[.663,.737]
4	0	1	...	1	.73	[.694,.766]
5	0	1	...	1	.69	[.653,.727]
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
1 mil	0	1	...	1	.74	[.705,.775]
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

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95% of the rows in this sampling distribution will have a CI that covers .70.

Why do the distributions have the same shape?

Informal logic:

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4. The variance of the sampling distribution of the proportion is $\sqrt{.7(1 - .7)/625} = .0183$.
5. The variance of the bootstrapped sampling distribution of the proportion is $\sqrt{.68(1 - .68)/625} = .0187$.

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3. The variance fully characterizes the shape of a normal distribution (not the location).
4. The variance of the sampling distribution of the proportion is $\sqrt{.7(1 - .7)/625} = .0183$.
5. The variance of the bootstrapped sampling distribution of the proportion is $\sqrt{.68(1 - .68)/625} = .0187$.
6. When n is large, these variances are close.

Two Questions

1. .68 is close to .70, did we get lucky? No! In a million runs, almost all within .05.
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Two Questions

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Summary of Population Inference for Proportions



Outline

Population Inference for Proportions

Population Inference for Means

The Current Population Survey

Something about the CPS sampling here...

Recall that our sample mean based on 2,258 working age Californians is \$135,406

Two Questions

Unlike the turnout example, we will never know the true population income mean, which we will denote with μ .

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1. Is \$135,406 close to the population mean μ ?
2. Can say something about how close?

Sampling Distribution for the Sample Mean

i s	1 Y_1	2 Y_2	...	2258 Y_{2258}	$\bar{Y}_{2258}$
1	37,006	108,811	...	49,001	135,406
$f.(.)$	?	?	...	?	?
$E.(.)$	μ	μ	...	μ	?
$V.(.)$	σ^2	σ^2	...	σ^2	?

Sampling Distribution for the Sample Mean

i	1	2	...	2258	
s	Y_1	Y_2	...	Y_{2258}	$\bar{Y}_{2258}$
1	37,006	108,811	...	49,001	135,406
2	152,001	174,261	...	75,105	135,956
3	4,800	52,000	...	169,900	133,790
4	$\vdots$	$\vdots$	...	$\vdots$	$\vdots$
5	$\vdots$	$\vdots$	...	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
1 mil	80,149	36,001	...	187,078	134,127
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$f(\cdot)$	?	?	...	?	?
$E(\cdot)$	μ	μ	...	μ	?
$V(\cdot)$	σ^2	σ^2	...	σ^2	?

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$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$f(\cdot)$	?	?	...	?	?
$E(\cdot)$	μ	μ	...	μ	?
$V(\cdot)$	σ^2	σ^2	...	σ^2	?

Property #1: *The sample mean is unbiased for the population mean.* The expected value of the sample mean, $E[\bar{Y}] = \mu$.

Proof:

$$\begin{aligned} E[\bar{Y}] &= E\left[\frac{1}{n}(Y_1 + \dots + Y_n)\right] \\ &= \\ &= \\ &= \\ &= \\ &= \mu. \end{aligned}$$

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$f(\cdot)$	?	?	...	?	?
$E(\cdot)$	μ	μ	...	μ	μ
$V(\cdot)$	σ^2	σ^2	...	σ^2	?

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Unlike the turnout example, we will never know the true population mean, which we will denote with μ . We will also never know the population standard deviation, which we will denote σ and the population variance with σ^2 .

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2. Can say something about how close?

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$f.(.)$	?	?	...	?	?
$E.(.)$	μ	μ	...	μ	μ
$V.(.)$	σ^2	σ^2	...	σ^2	?

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2	152,001	174,261	...	75,105	135,956
3	4,800	52,000	...	169,900	133,790
4	$\vdots$	$\vdots$	...	$\vdots$	$\vdots$
5	$\vdots$	$\vdots$	...	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
1 mil	80,149	36,001	...	187,078	134,127
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$f(\cdot)$	?	?	...	?	?
$E(\cdot)$	μ	μ	...	μ	μ
$V(\cdot)$	σ^2	σ^2	...	σ^2	?

Sampling Distribution for the Sample Mean

i	1	2	...	2258	
s	Y_1	Y_2	...	Y_{2258}	$\bar{Y}_{2258}$
1	37,006	108,811	...	49,001	135,406
2	152,001	174,261	...	75,105	135,956
3	4,800	52,000	...	169,900	133,790
4	$\vdots$	$\vdots$	...	$\vdots$	$\vdots$
5	$\vdots$	$\vdots$	...	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
1 mil	80,149	36,001	...	187,078	134,127
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$f(\cdot)$	?	?	...	?	?
$E(\cdot)$	μ	μ	...	μ	μ
$V(\cdot)$	σ^2	σ^2	...	σ^2	?

Property #2: The *sampling variance* of the sample mean,

$$V[\bar{Y}] = \frac{\sigma^2}{n}$$

Proof:

$$\begin{aligned} V[\bar{Y}] &= V \left[\left(\frac{1}{n} Y_1 + \dots + \frac{1}{n} Y_n \right) \right] \\ &= \\ &= \\ &= \\ &= \\ &= \frac{1}{n}. \end{aligned}$$